Ethische Aspekte neuer Technologien

An Overview and Introduction

Lecturer: Lukas Madl

06.11.26

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What constitutes a Good Life?

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Technology Ethics?

Where does digital ethics come from – A historic account

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Impacts get long-term

Hans Jonas

1903 to 1993

Philosopher, Theologian

Responsibility for

Long-term impact



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Das Prinzip Verantwortung

„Der endgültig entfesselte Prometheus, dem die Wissenschaft nie gekannte Kräfte und die Wirtschaft den rastlosen Antrieb gibt,
ruft nach einer Ethik, die durch freiwillige Zügel seine Macht davor zurückhält, dem Menschen zum Unheil zu werden“



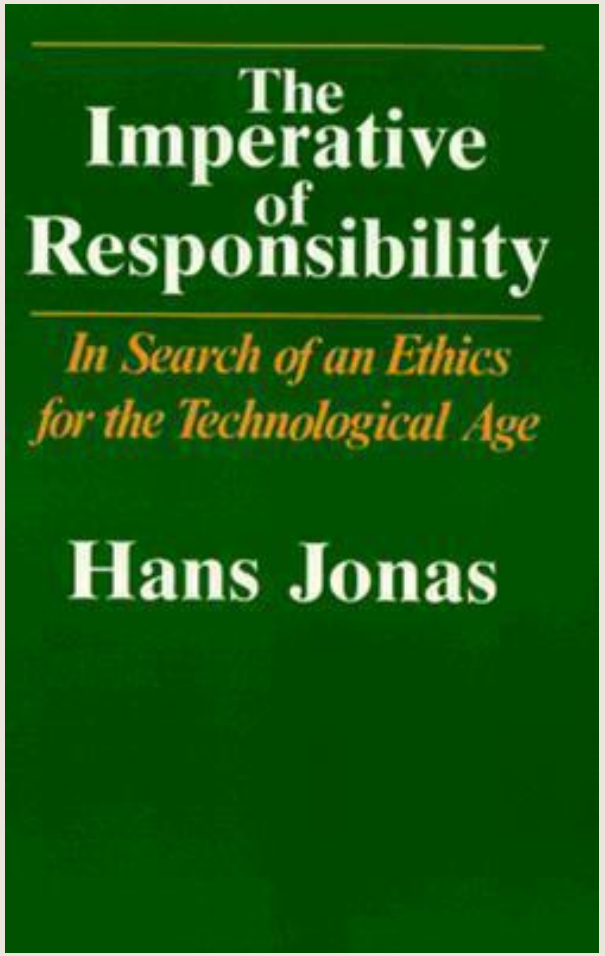
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BAD
HERE & NOW
GOOD
RESPONSIBILITY
FUTURE
N
W
E
S
UNTIL NOW
CHARACTERISTICS OF PREVIOUS ETHICS
1o INTERACTION BETWEEN PEOPLE
2o STATE OF HUMANS
3o EFFECT CONSTANT
IS CLOSE
HOMO SAPIENS
homo faber
TECHNE
NOW
HUMAN AS OBJECT
TIME IMPACT
HOMO FABER
ETHICS
NO
YES responsibility
NO
NATURE
SPACE IMPACT
NOT KNOWING


NEURALINK

“Act so that the effects of your action are compatible with the permanence of genuine human life on Earth.”

„Handle so, dass die Wirkungen deiner Handlung verträglich sind mit der Permanenz echten menschlichen Lebens auf Erden.“



Hans Jonas, The Imperative of Responsibility (1979)

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Exercise: Jonas #1

What do you think?

On a scale of 0 (not at all) to 10 (100 per cent):

How likely is it that AI will pose a genuine threat to human life over the next 20 years?

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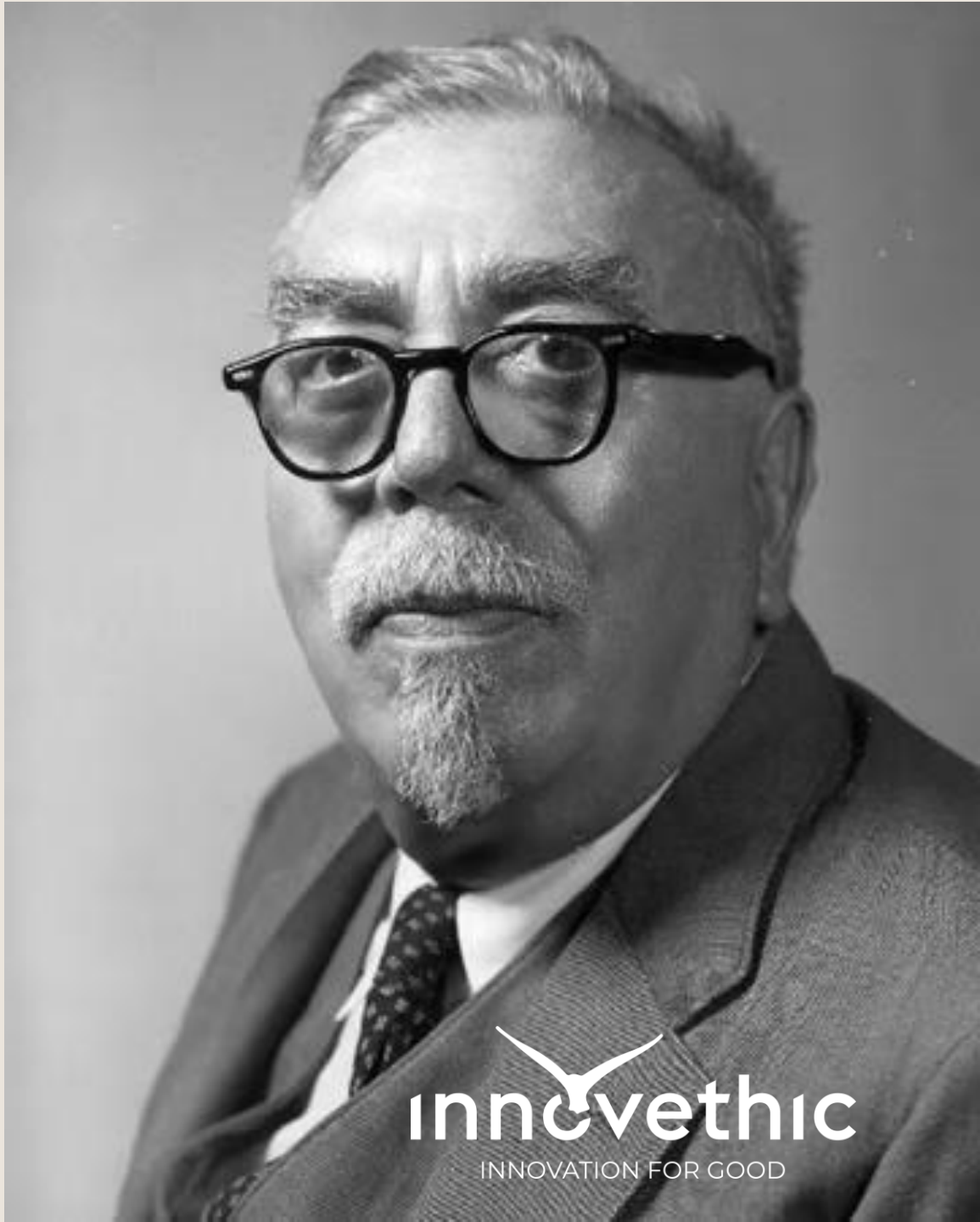
Machines get smart

Norbert Wiener

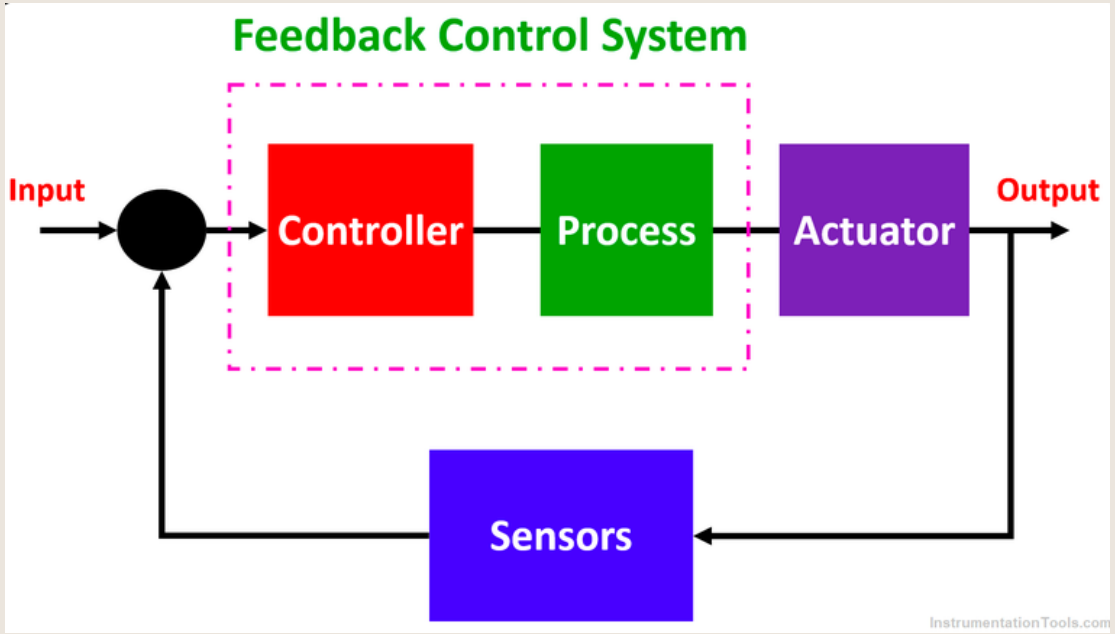
1894 to 1964

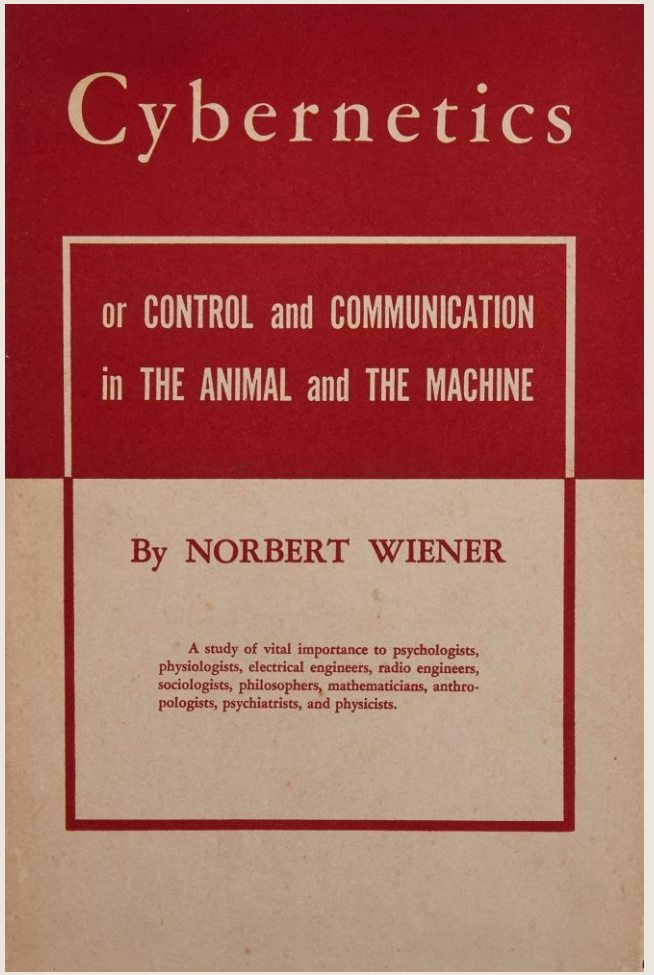
Computer Scientist, Mathematician, Philosopher

Originator of cybernetics



Cybernetics



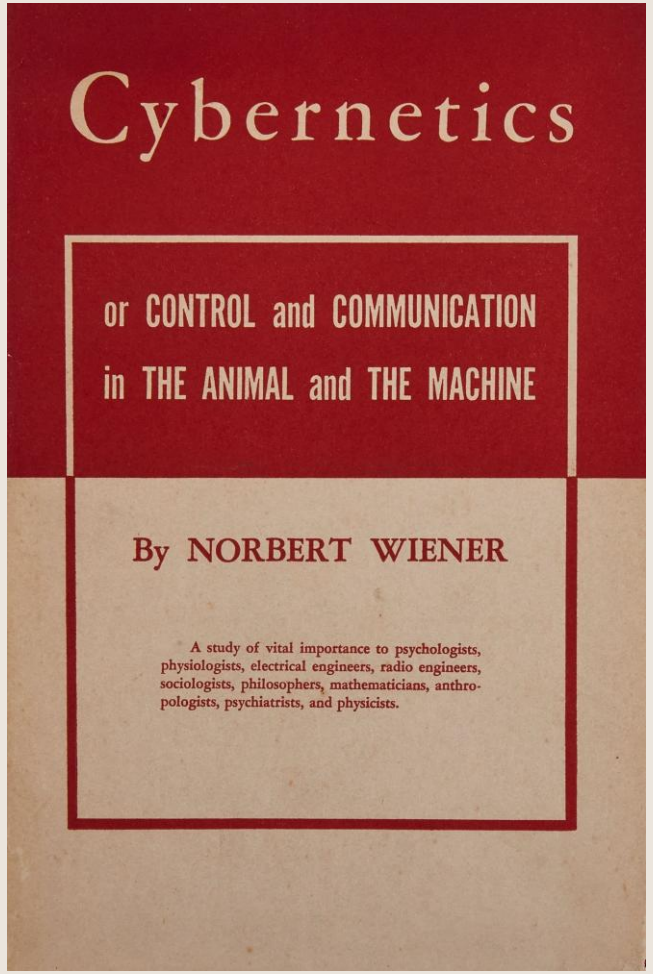


intellectual foundations for modern **computer science**, systems theory, **robotics**, **artificial intelligence**, and neuroscience.

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“we are already in a position to construct artificial machines of almost any degree of elaborateness of performance. …

we were here in the presence of another social potentiality of unheard-of importance for good and for evil.”



Norbert Wiener, Cybernetics or Control and Communication in the Animal and the Machine (1948)

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Exercise: Wiener #1

Give an example of an AI application with

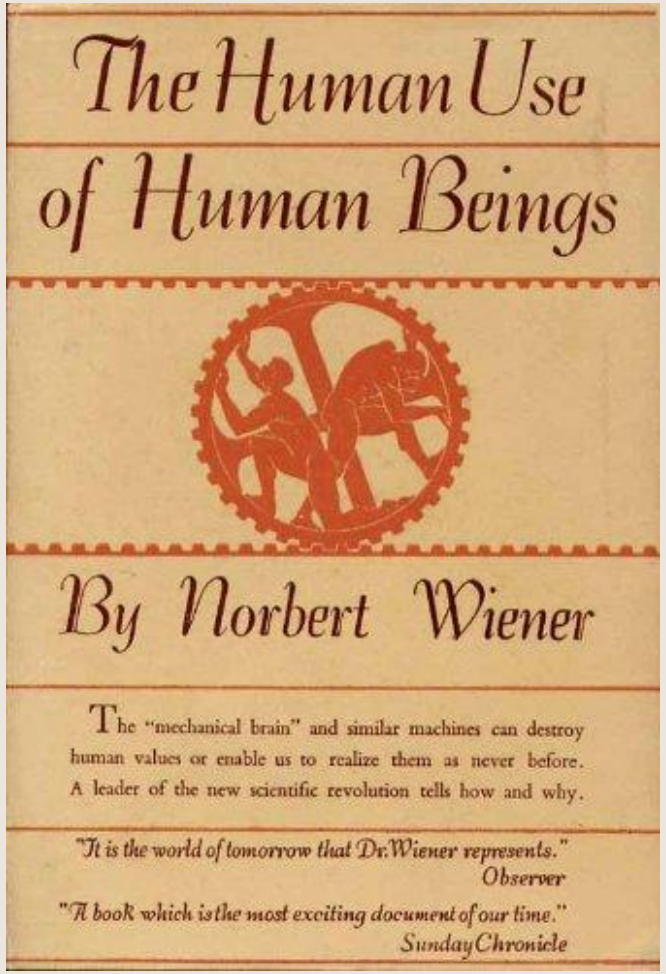
a. positive effects

b. and one with negative (harmful) effects

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„The machine's danger to society is not from the machine itself but from what man makes of it."



Norbert Wiener, The Human Use of Human Beings (1950)

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The Human Use of Human Beings

Key topics

- Automation and labor
- Responsibility and control
- Human machines symbiosis

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Exercise: Wiener #2

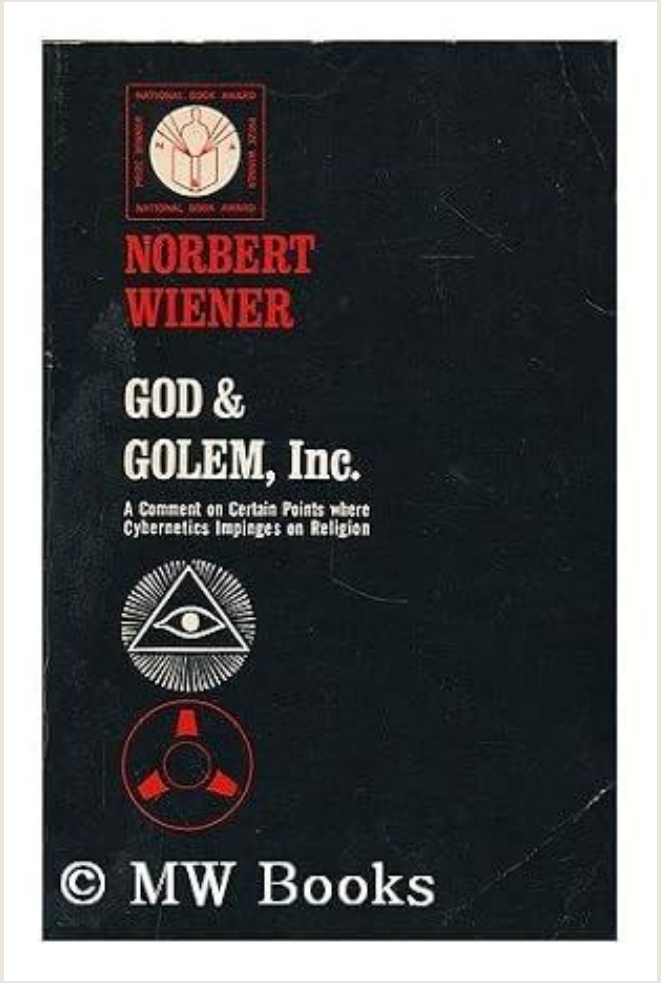
a. Give me one example of a vocational qualification that you think will soon be replaced by AI.
b. Give me one example of a decision that should not be made by a machine.
c. List one characteristic that humans and machines (AI) have in common, and one characteristic that only humans have.

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“Render unto man the things which are man's and unto the computer the things which are the computer's.”

Norbert Wiener, God & Golem (1964)



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Machines get human-like?

Joseph Weizenbaum

1923 to 2008

Computer Scientist

Developed ELIZA



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ELIZA, 1966



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*“The Internet is like one of those **garbage** dumps outside of Bombay. There are people, most unfortunately, crawling all over it, and maybe they find a bit of aluminum, or perhaps something they can sell. But mainly it’s garbage.”*

Joseph Weizenbaum, New York Times (1999)

*“There are gold mines and **pearls** in there that a person trained to design good questions can find.”*

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Exercise: Weizenbaum #1

Write down: Give one specific example of

a) 'rubbish' on the internet, and

b) one specific example of a 'pearl'.

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Weizenbaum made crucial distinction

Deciding

Computational process, calculated, automated, based on formal rules

Choosing

Human act. Involves judgment, values and responsibility.

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Exercise: Weizenbaum #2

Write down:

a. What factors do you take into account when choosing who to go on holiday with?

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Exercise: Weizenbaum #3

Write down one example each:

Which decisions would you

a) readily leave to an algorithm, and

b) which should remain a human choice?

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Computers are malleable

James Moor

1942 to 2024

Philosopher

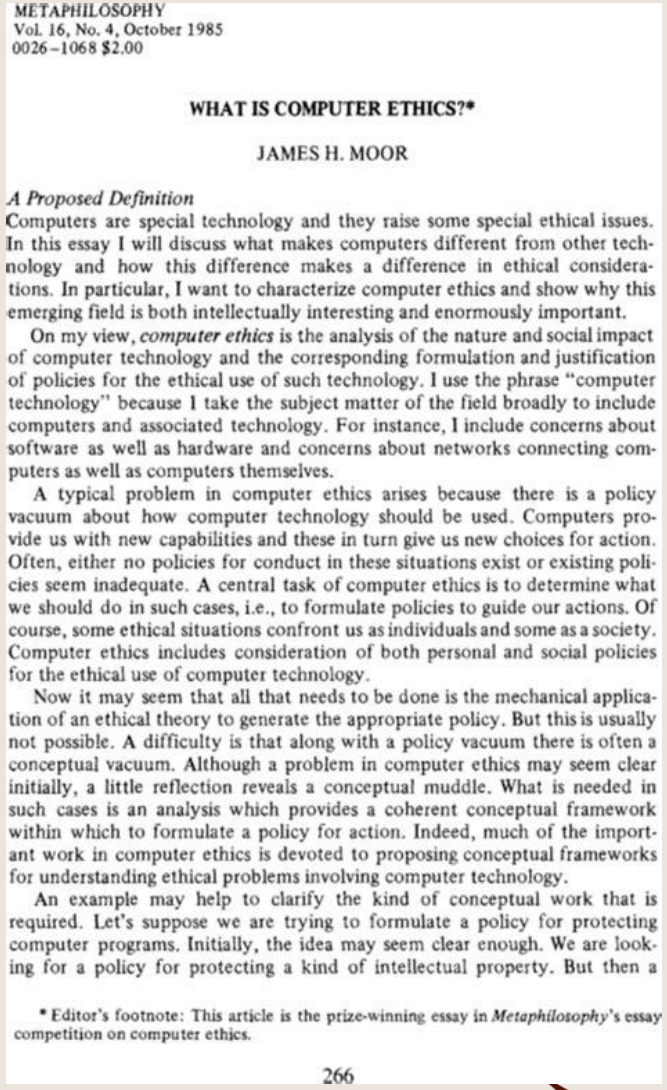
Pioneer of computer ethics



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"The revolutionary feature of computers is their logical malleability. Logical malleability assures the enormous application of computer technology. This will bring about the Computer Revolution. During the Computer Revolution many of our human activities and social institutions will be transformed. These transformations will leave us with policy and conceptual vacuums about how to use computer technology."

James Moor, What is computer Ethics? (1985)



METAPHILOSOPHY
Vol. 16, No. 4, October 1985
0026–1068 $2.00

WHAT IS COMPUTER ETHICS?*

JAMES H. MOOR

A Proposed Definition

Computers are special technology and they raise some special ethical issues. In this essay I will discuss what makes computers different from other technology and how this difference makes a difference in ethical considerations. In particular, I want to characterize computer ethics and show why this emerging field is both intellectually interesting and enormously important.

On my view, *computer ethics* is the analysis of the nature and social impact of computer technology and the corresponding formulation and justification of policies for the ethical use of such technology. I use the phrase "computer technology" because I take the subject matter of the field broadly to include computers and associated technology. For instance, I include concerns about software as well as hardware and concerns about networks connecting computers as well as computers themselves.

A typical problem in computer ethics arises because there is a policy vacuum about how computer technology should be used. Computers provide us with new capabilities and these in turn give us new choices for action. Often, either no policies for conduct in these situations exist or existing policies seem inadequate. A central task of computer ethics is to determine what we should do in such cases, i.e., to formulate policies to guide our actions. Of course, some ethical situations confront us as individuals and some as a society. Computer ethics includes consideration of both personal and social policies for the ethical use of computer technology.

Now it may seem that all that needs to be done is the mechanical application of an ethical theory to generate the appropriate policy. But this is usually not possible. A difficulty is that along with a policy vacuum there is often a conceptual vacuum. Although a problem in computer ethics may seem clear initially, a little reflection reveals a conceptual muddle. What is needed in such cases is an analysis which provides a coherent conceptual framework within which to formulate a policy for action. Indeed, much of the important work in computer ethics is devoted to proposing conceptual frameworks for understanding ethical problems involving computer technology.

An example may help to clarify the kind of conceptual work that is required. Let's suppose we are trying to formulate a policy for protecting computer programs. Initially, the idea may seem clear enough. We are looking for a policy for protecting a kind of intellectual property. But then a

* Editor's footnote: This article is the prize-winning essay in *Metaphilosophy*'s essay competition on computer ethics.

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Exercise: Moor #1

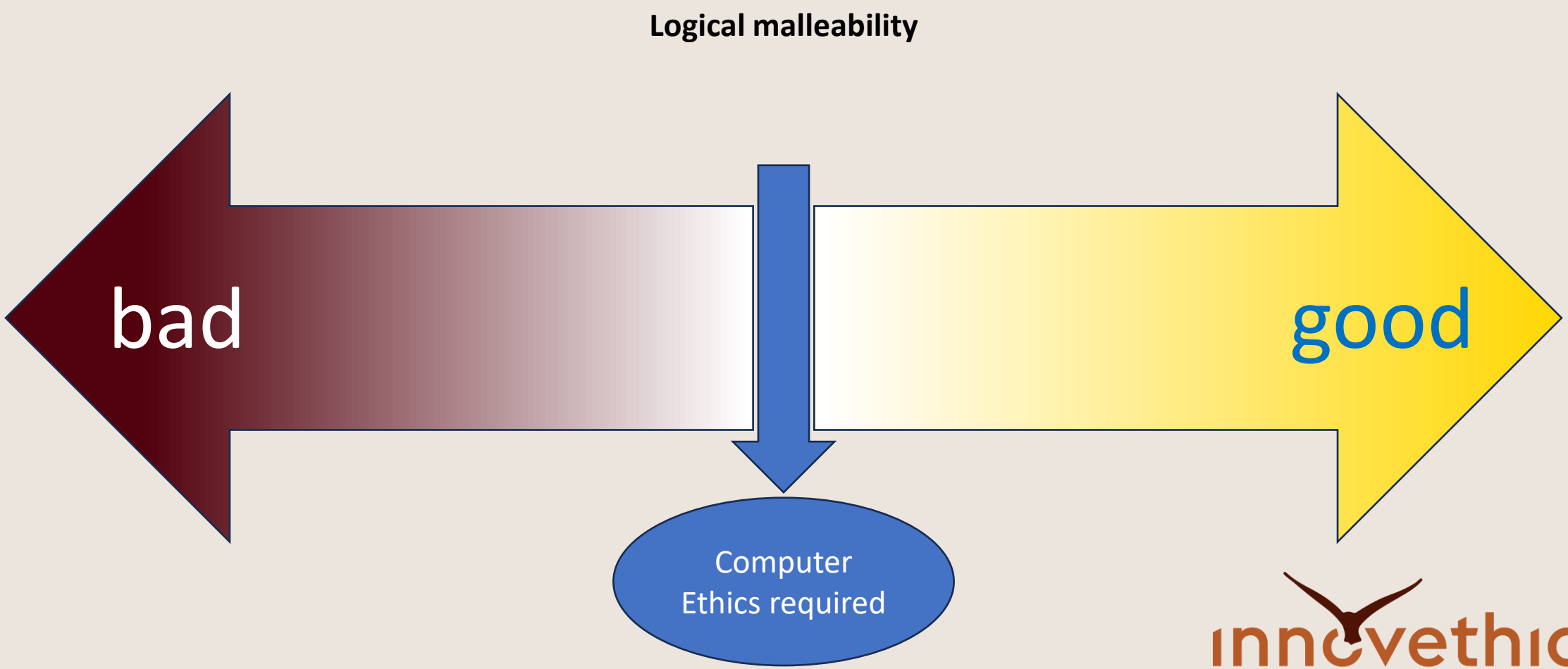
Give an example of a ‘political vacuum’ in the field of AI.

Where are rules needed on how AI applications should be used?

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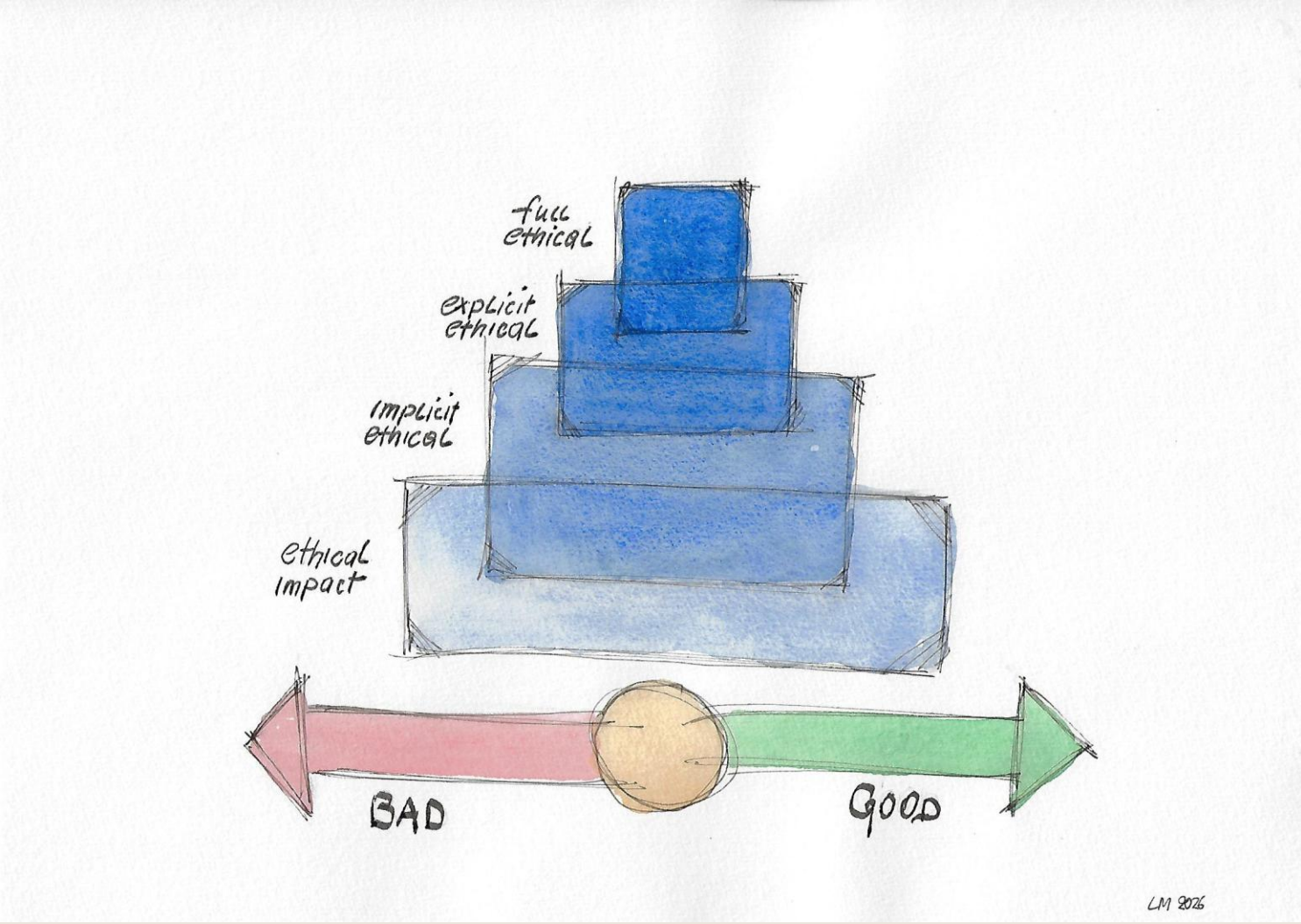
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James Moor: What's new about Computers?



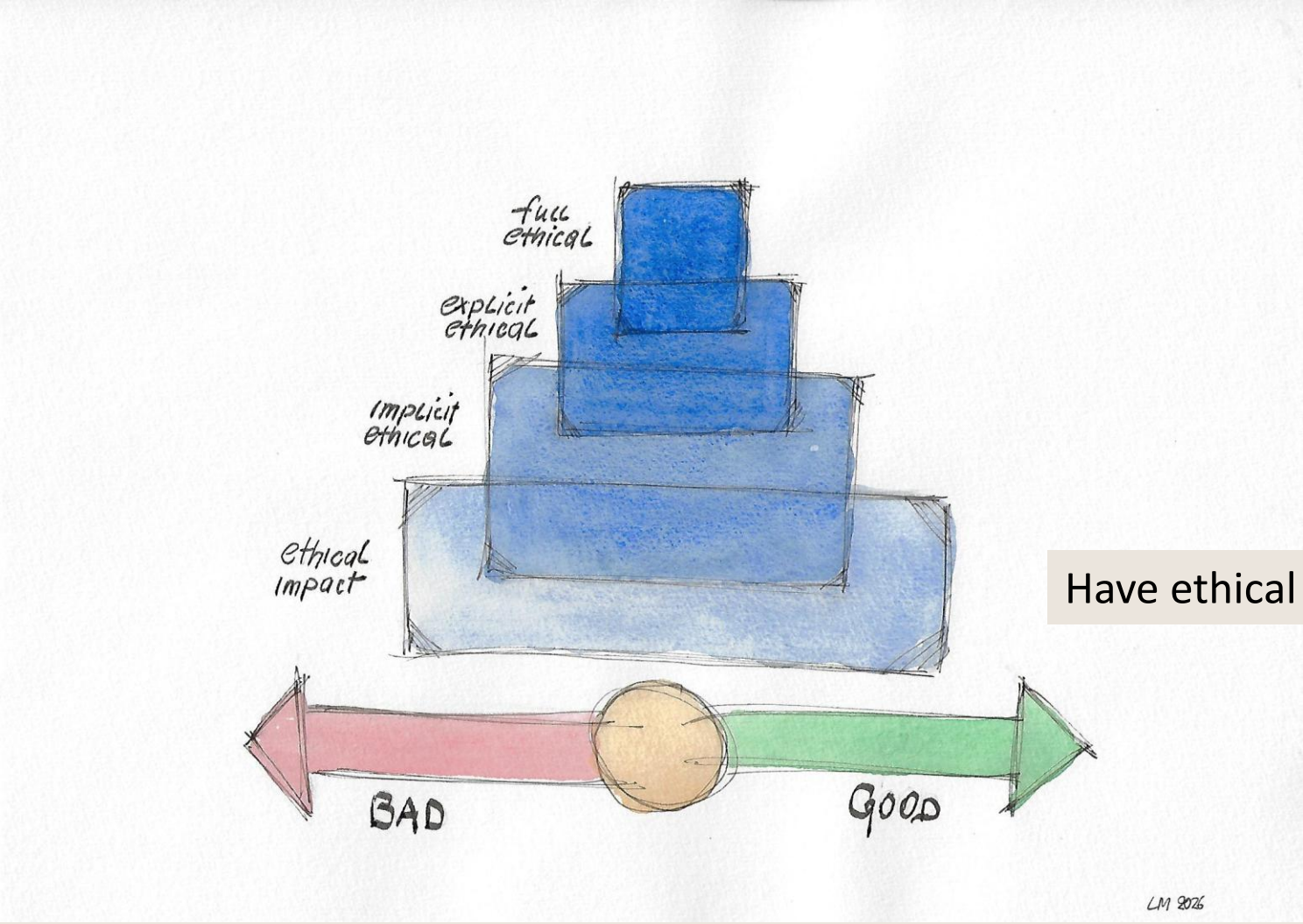
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4 Levels of Ethical Agency



„The nature, importance, and difficulty of machine ethics“ 2006

Robot Ethics



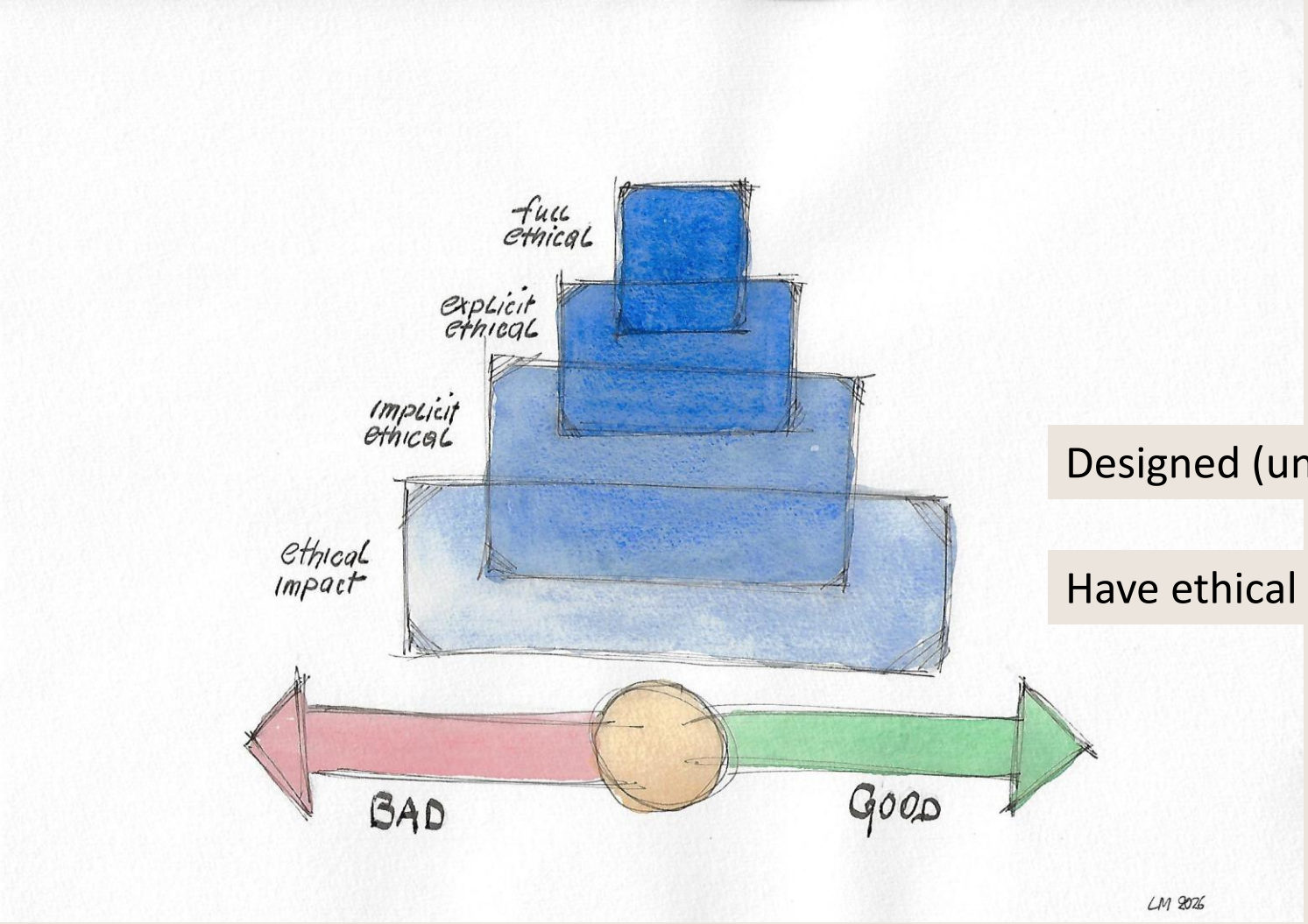
Have ethical effects

A historic example for negative ethical impact



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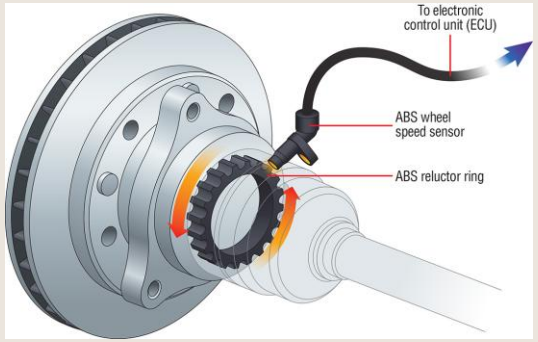
Robot Ethics



Designed (un)ethically

Have ethical effects

Implicit ethical agents

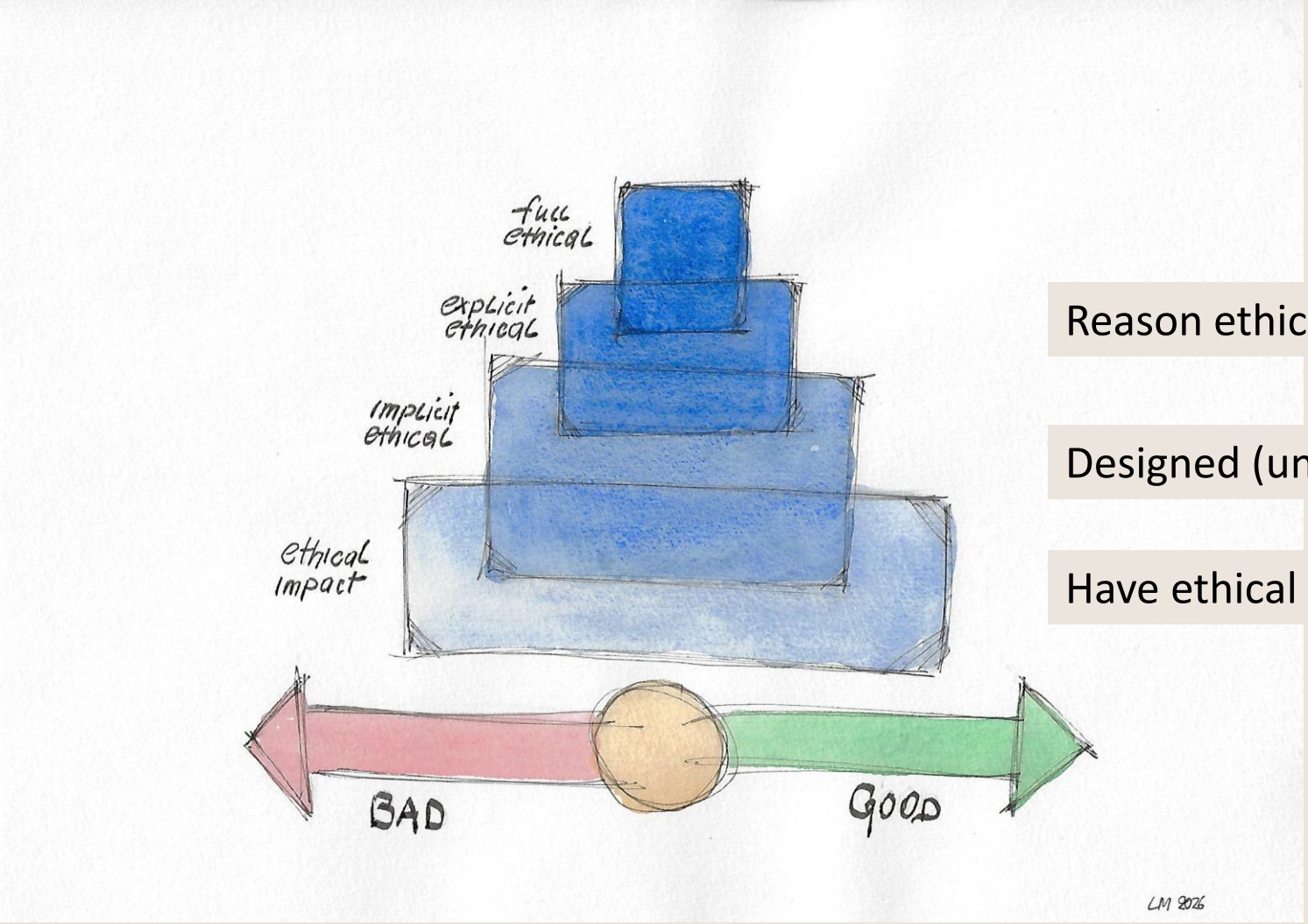






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Robot Ethics



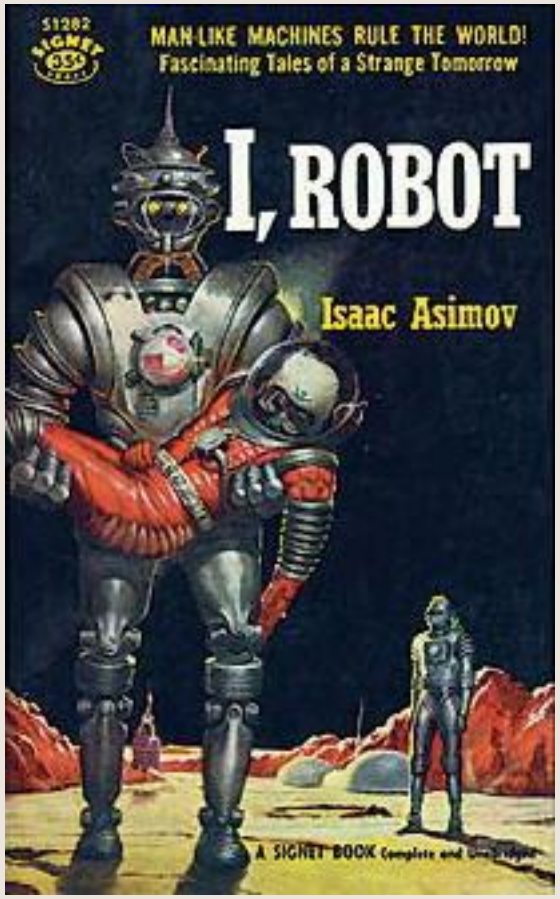
Reason ethically

Designed (un)ethically

Have ethical effects

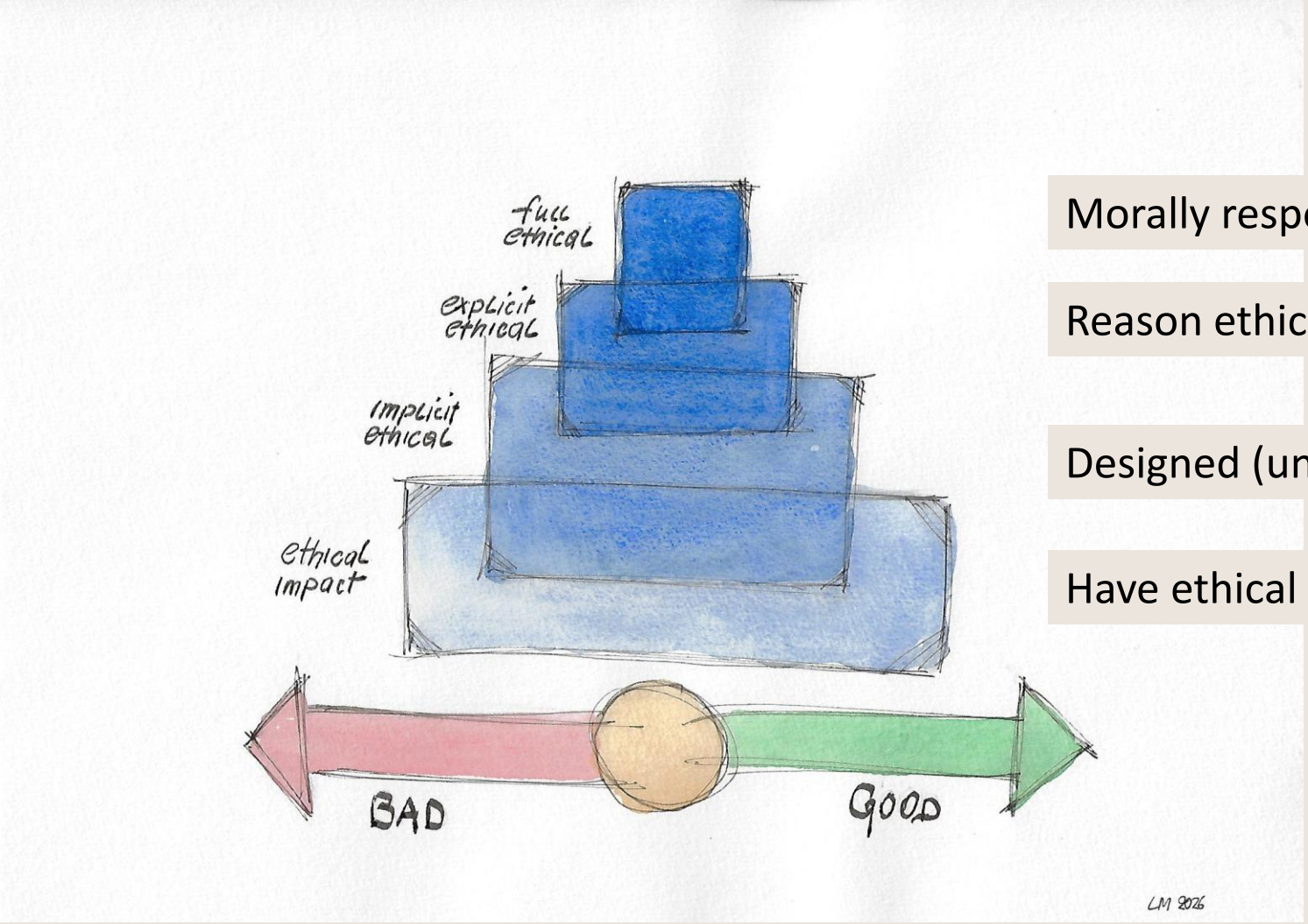
Explicit ethical agents

- A robot may not injure a human being or, through inaction, allow a human being to come to harm.
- A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.
- A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.



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Robot Ethics



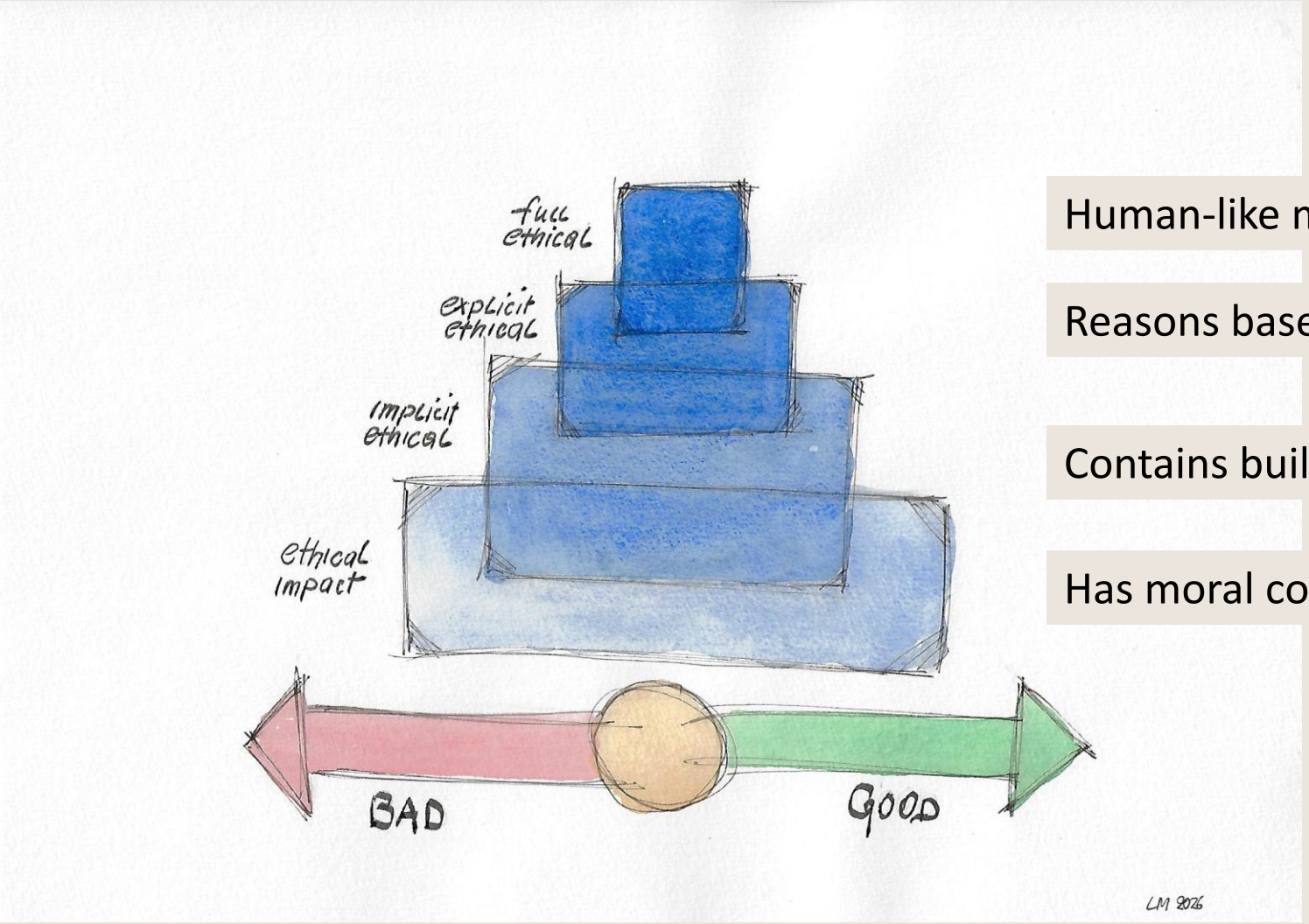
Morally responsible

Reason ethically

Designed (un)ethically

Have ethical effects

Robot Ethics



Human-like moral agency

Reasons based on ethical principles

Contains built-in ethical constraints

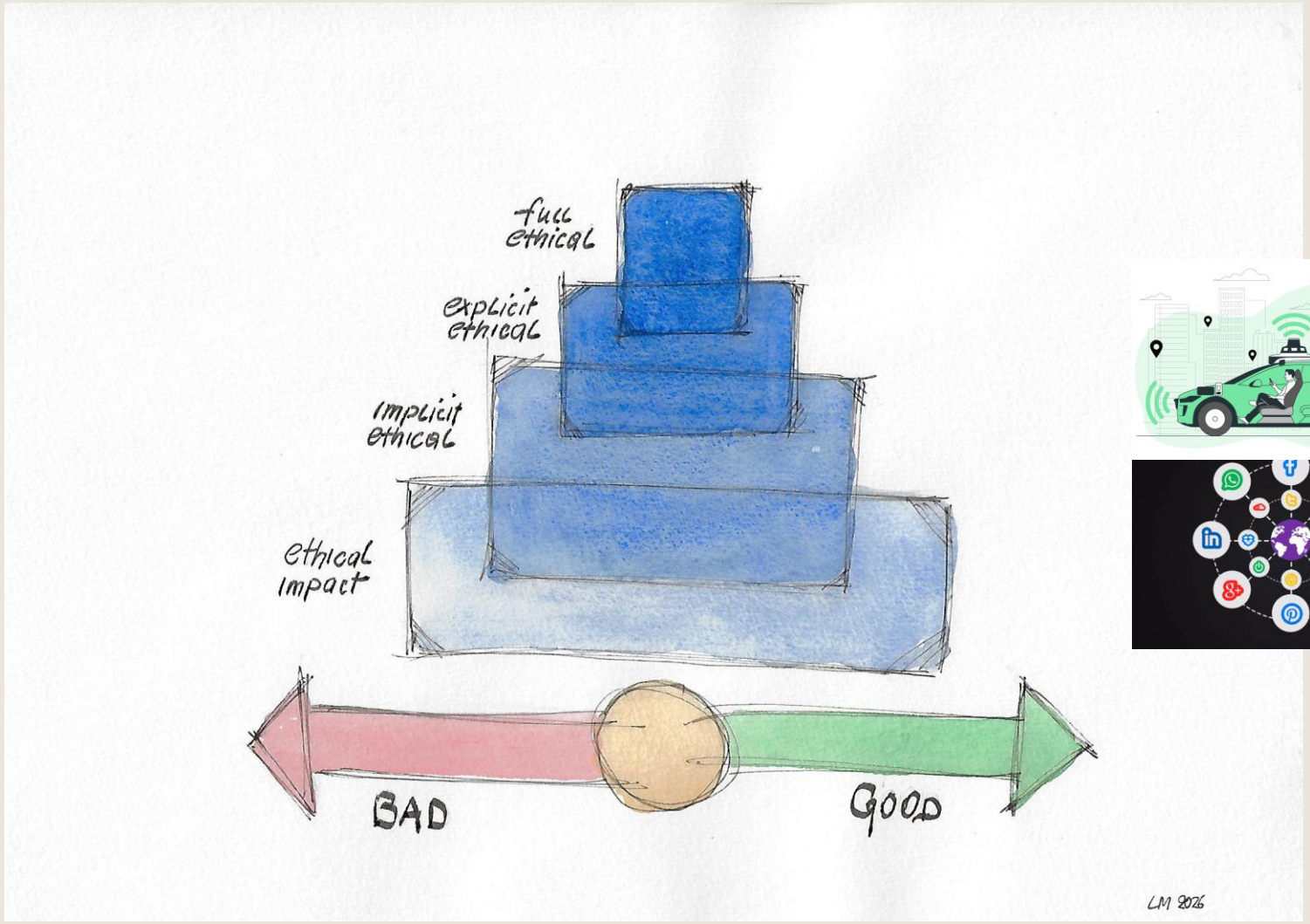
Has moral consequences

At a Glance

Type	Can cause ethical impact	Ethics built in	Reason ethically	Has consciousness
Full ethical	✅	✅	✅	✅
Explicit ethical	✅	✅	✅	❌
Implicit ethical	✅	✅	❌	❌
Ethical impact	✅	❌	❌	❌

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Robot Ethics





Autonomous cars



Social Media

George Orwell

1903 to 1950

Writer and journalist

Dystopian surveillance novel



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Relevant topics in the novel

- Constant surveillance
- Power and information control
- Behavioural manipulation
- Language control

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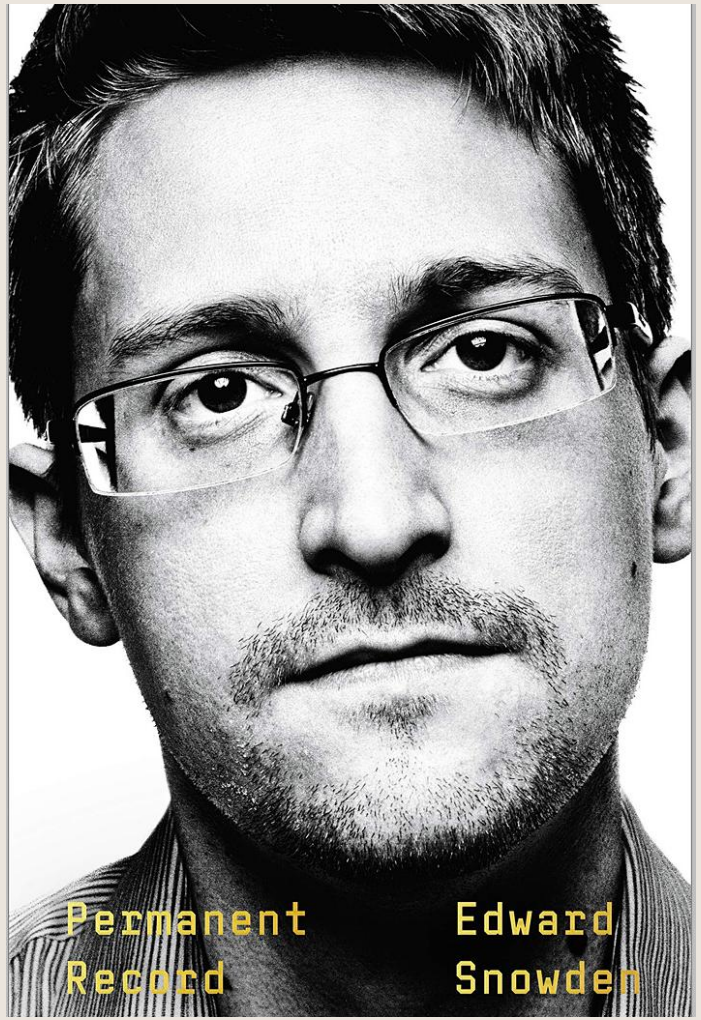
Edward Snowden

*1983

Former contractor at NSA

Most influental whistle-blower

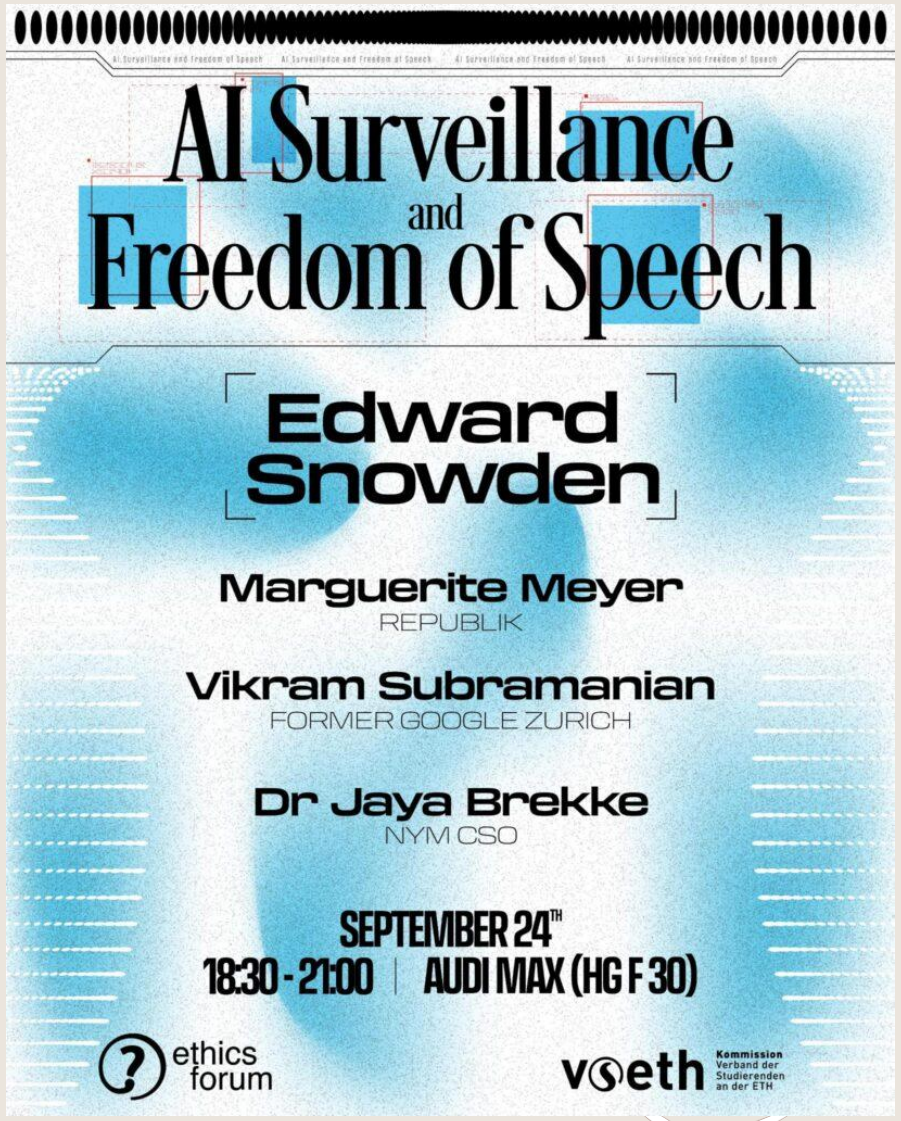
2013: disclosure of large collection of confidential documents



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“I think we need to put Sam Altmann in prison“

Edward Snowden, ETHI Zürich conference, 2026



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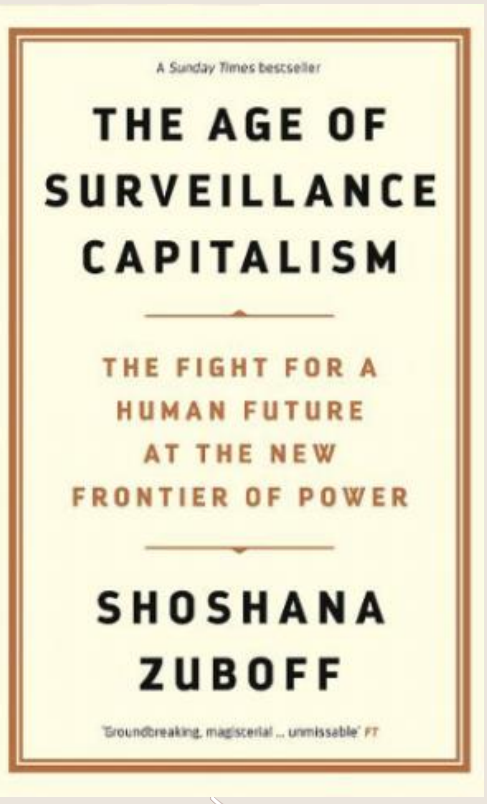
Human experience into raw material for profit

Shashana Zuboff

*1951

author, professor,
social psychologist,
philosopher,





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Surveillance Capitalism

A new economic system where companies collect, analyze, and profit from personal data—often without users' full awareness or consent.

Data extraction:

- Companies (especially tech giants) gather vast amounts of behavioral data from users.

Prediction products:

- This data is used to predict and influence future behavior.

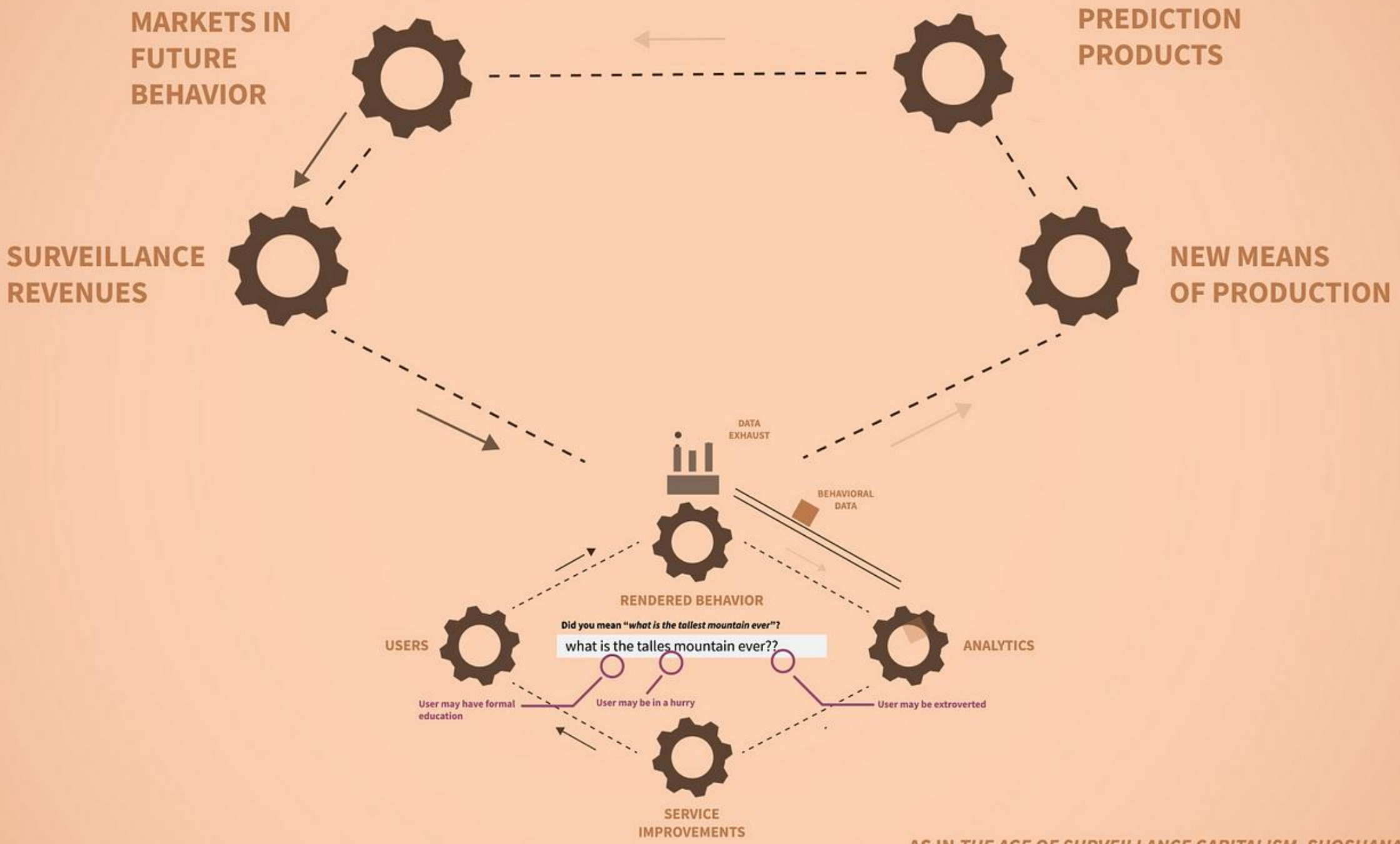
Profit motive:

- These predictions are sold to advertisers and other third parties for profit.

Loss of autonomy:

- It shifts power from individuals to corporations, eroding privacy and personal freedom.
- Zuboff argues that this model turns human experience into a raw material for commercial practices, fundamentally changing the relationship between individuals and the digital world.

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AS IN *THE AGE OF SURVEILLANCE CAPITALISM, SHOSHANA ZUBOFF*

Example: The social dilemma



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„I am tired of feeling like I‘m dumb,
not pretty enough,
not rich enough,
not popular enough,
not everything enough“

Lilly, 20, Los Angeles, USA

Source: https://socialtruth.humanetech.com/

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ROGER MCNAMEE

EARLY FACEBOOK INVESTOR; AUTHOR OF "ZUCKED: WAKING UP TO THE FACEBOOK CATASTROPHE"



JEFF SEIBERT

FORMER SENIOR DIRECTOR OF PRODUCT AT TWITTER



SHOSHANA ZUBOFF

PROFESSOR EMERITUS AT HARVARD BUSINESS SCHOOL; AUTHOR OF "THE AGE OF SURVEILLANCE CAPITALISM"



JOE TOSCANO

FORMER EXPERIENCE DESIGN CONSULTANT AT GOOGLE; AUTHOR OF "AUTOMATING HUMANITY"



JUSTIN ROSENSTEIN

CO-FOUNDER OF ASANA AND ONE PROJECT; FORMER ENGINEERING LEAD AT FACEBOOK; FORMER PRODUCT MANAGER AT GOOGLE



TIM KENDALL

CEO OF MOMENT; FORMER DIRECTOR OF MONETIZATION AT FACEBOOK; FORMER PRESIDENT OF PINTEREST

The technology that **connects** us

Also

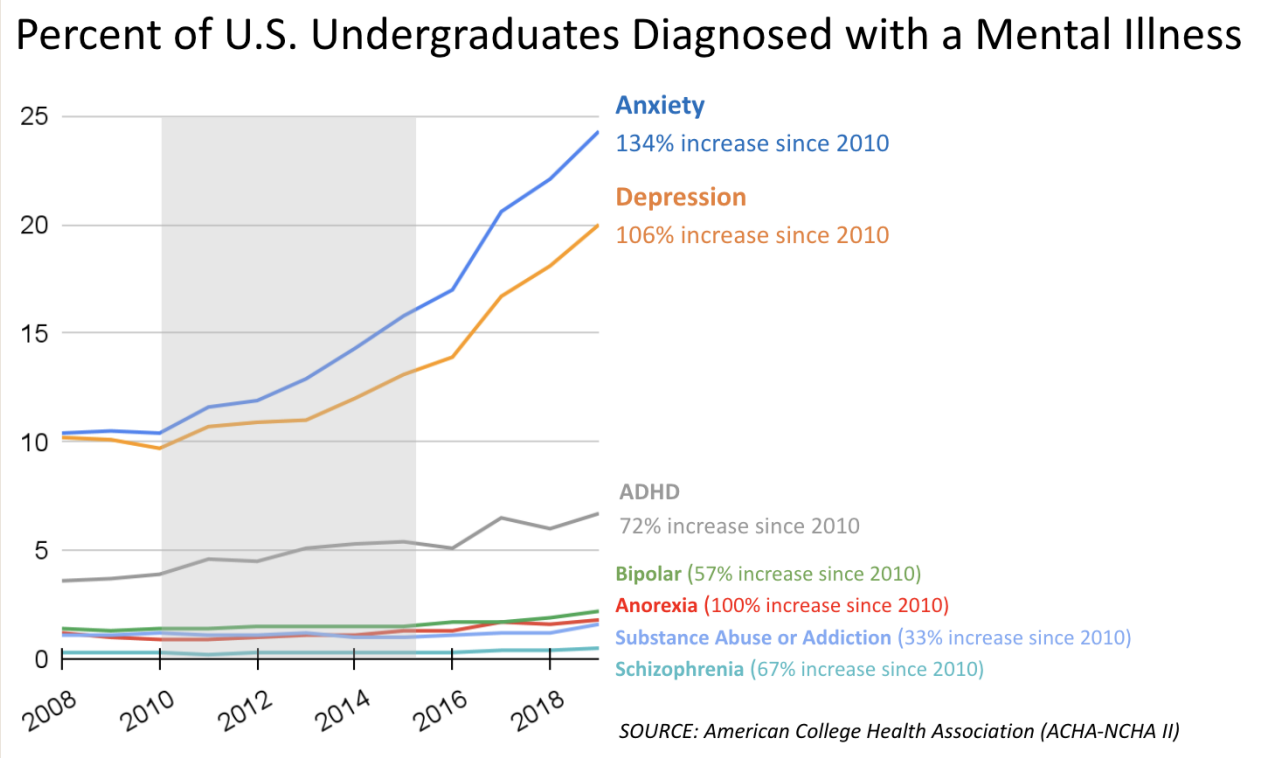
- **divides** us
- **controls** us
- **manipulates** us
- **monetizes** us

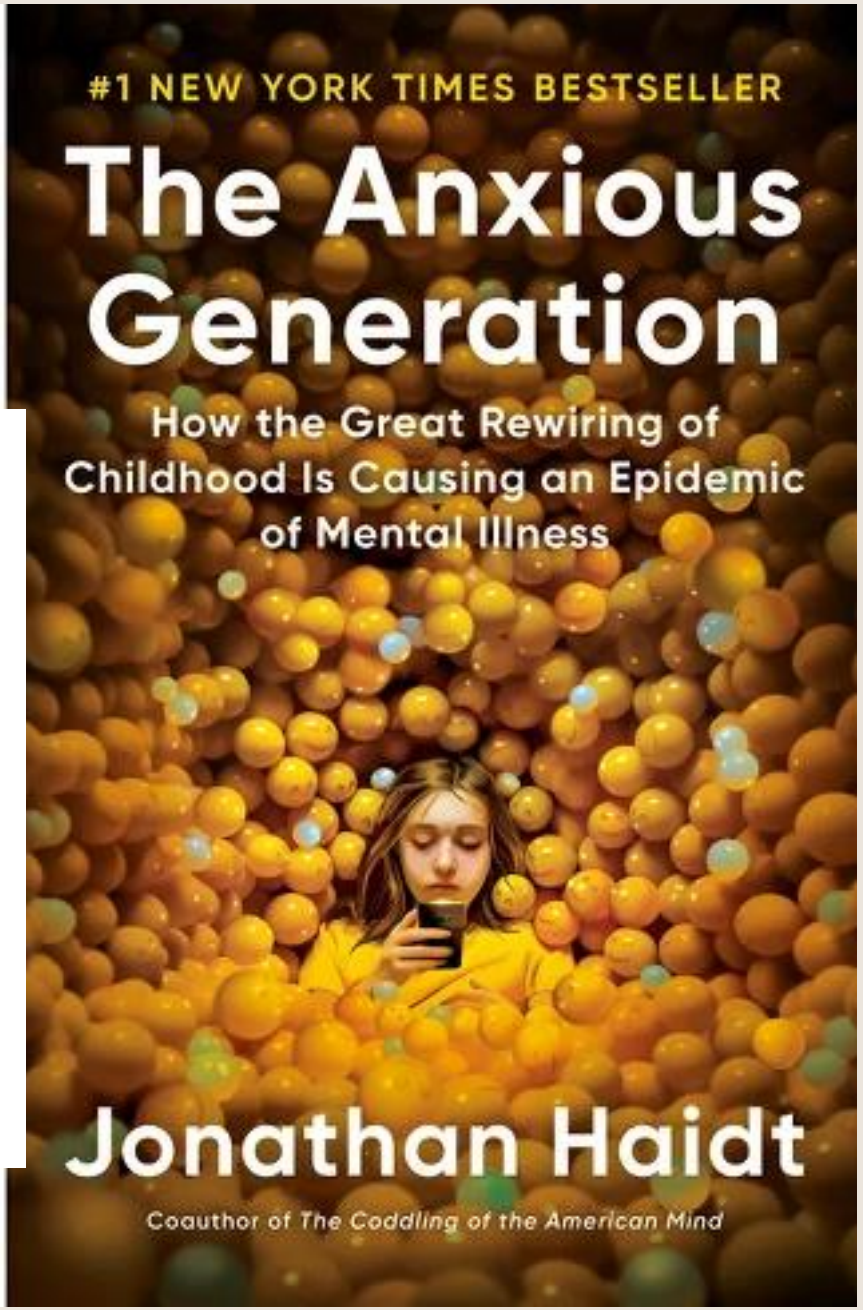
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That is a problem:

- For mental Health





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That is a problem:

- For mental Health
- For Democracy

The # of countries with political disinformation campaigns on social media doubled in the past two years.

the new York Times
2019

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That is a problem:

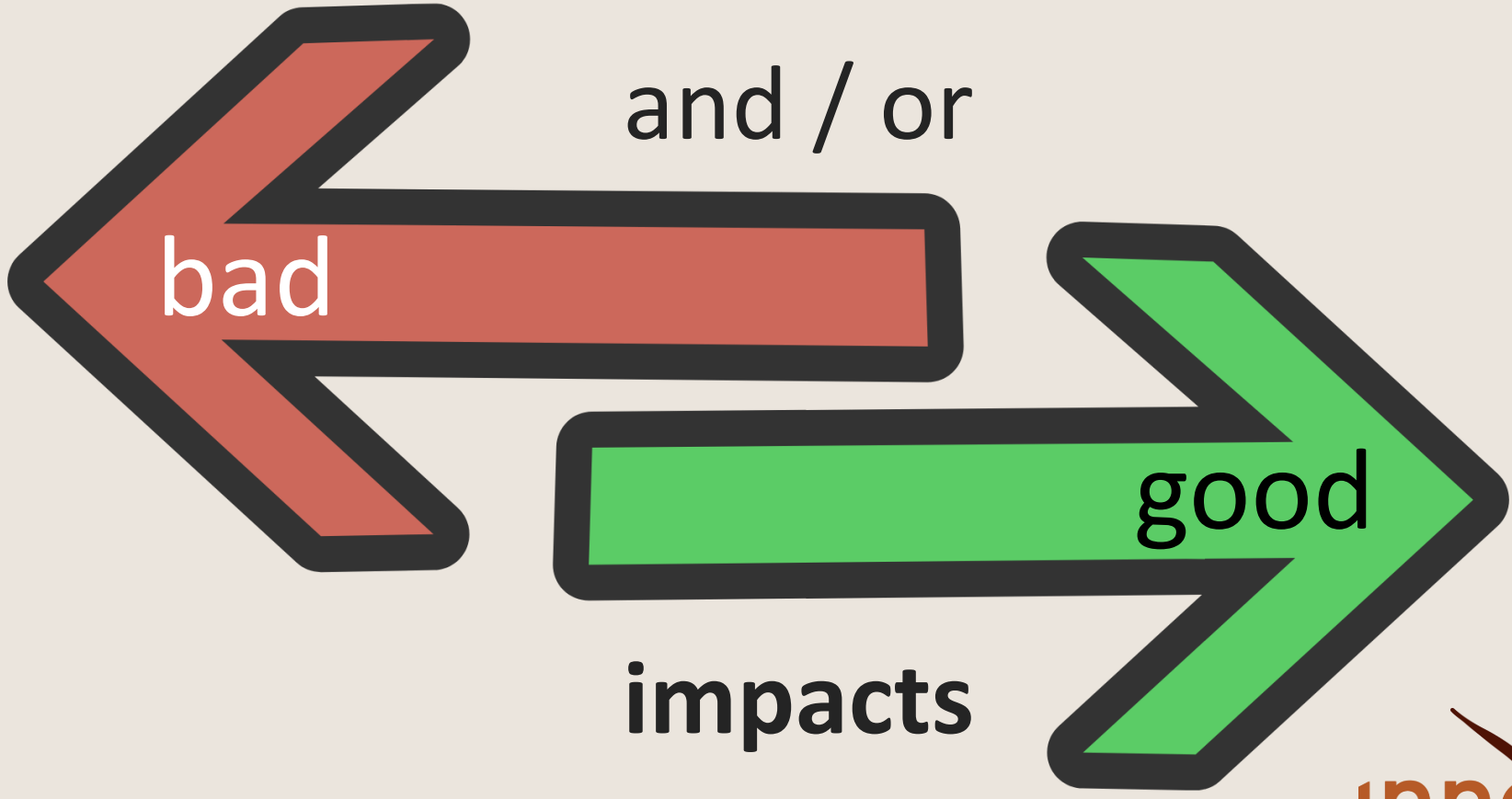
- For mental Health
- For Democracy
- For social cohesion



64% of the people who joined extremist groups on Facebook did so because the algorithms steered them there.

Internal Facebook report 2018

Technology can have



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How can a qualitative distinction be made between good and bad effects?

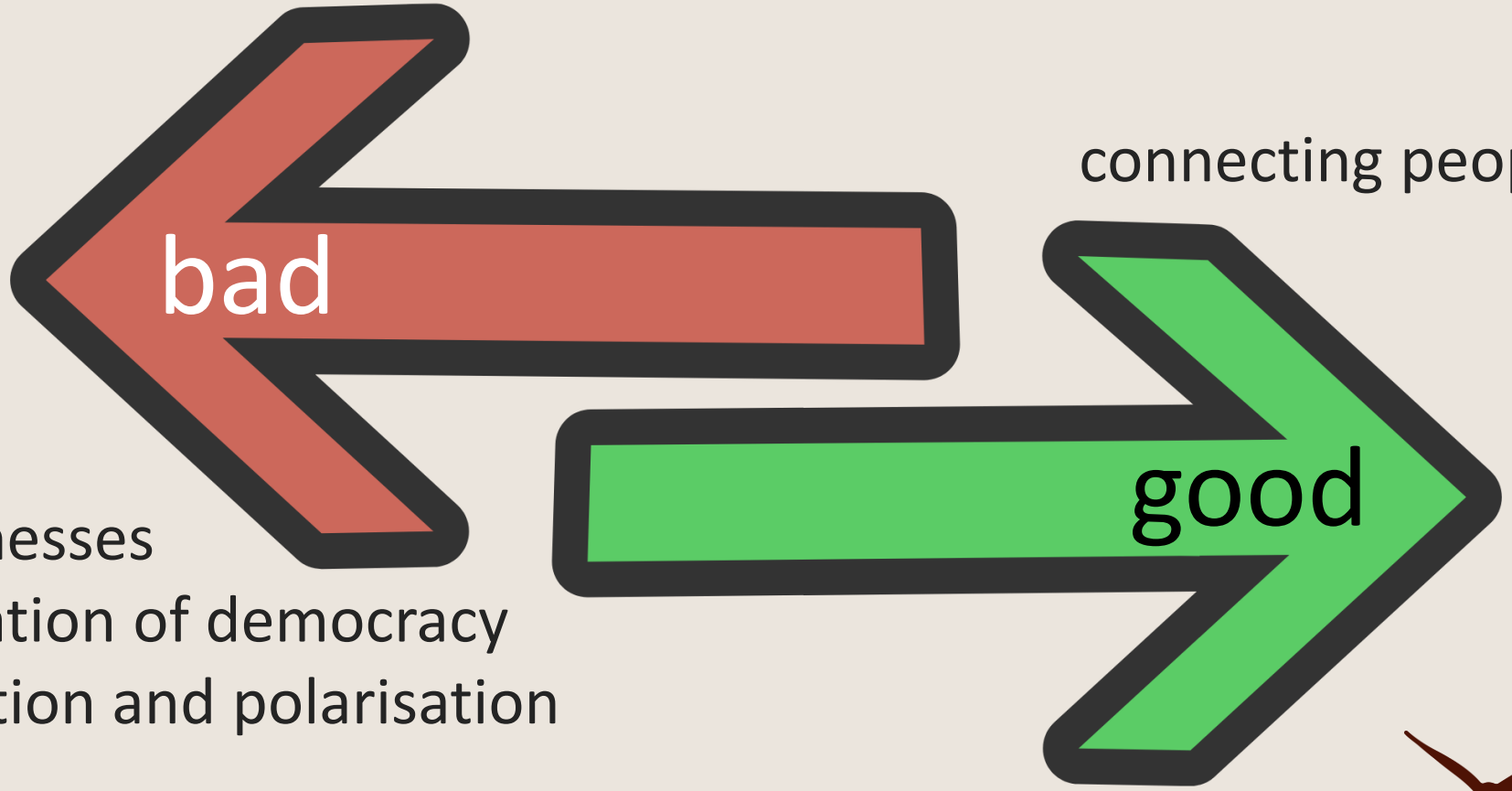
That’s the purpose of ethics:

In the broad sense,
ethics is the reflected investigation
of the question of what a ***good life*** means
and how it can be achieved.

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Social Media (as currently set up)

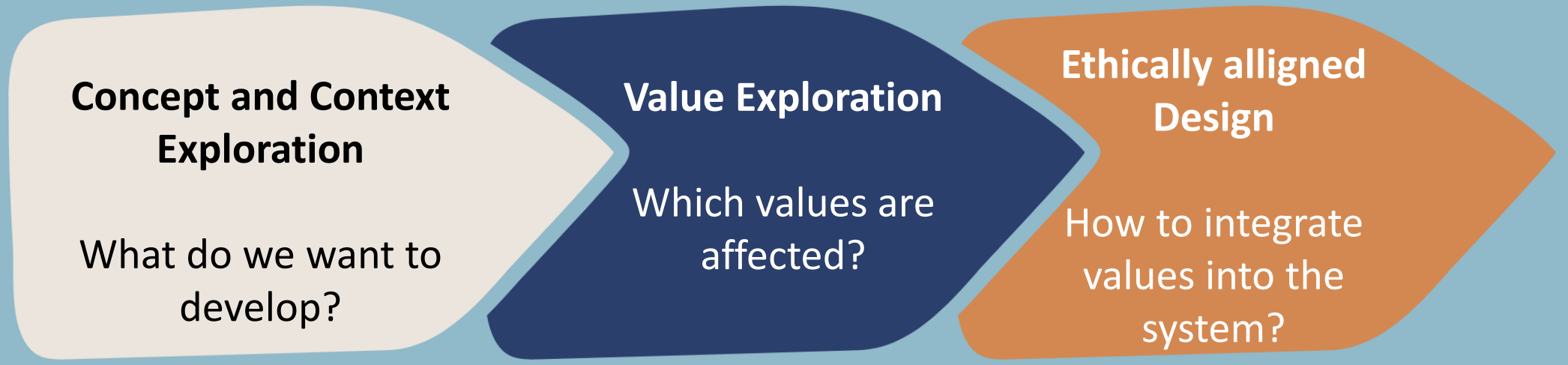


mental illnesses
destabilisation of democracy
extremisation and polarisation

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Value Based Engineering in three Phases



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Magnifica Humanitas

Pope Leo XIV

May 2026



Criteria in accordance with Christian social teaching.

- Dignity
- Common good
- General determination of goods
- Subsidiarity
- Solidarity
- Social justice

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WITH GREAT POWER
COMES GREAT
RESPONSIBILITY.
SPIDERMAN
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